39. What are near point and normal focusing?
40. Why is oil immersed objective preferred in a microscope?
41. What are the advantages and disadvantages of a reflecting telescope?
42. What is the use of an erecting lens in a terrestrial telescope?
43. What is the use of collimator in a spectrmeter?
44. What are the uses of spectrometer?
45. What is myopia? What is its remedy?
46. What is hypermetropia? What is its remedy?
47. What is astigmatism? What is its remedy?
48. What is presbyopia?

Long Answer Questions

1. Prove law of reflection using Huygens' principle.
2. Prove law of refraction using Huygens' principle.
3. Obtain the equation for resultant intensity due to interference of light.
4. Explain the Young's double slit experimental setup and obtain the equation for path difference.
5. Obtain the equation for bandwidth in Young's double slit experiment.
6. Discuss the interference in thin films and obtain the equations for constructive and destructive interference for transmitted and reflected light.
7. Discuss the diffraction at single slit and obtain the condition for n^{th} minimum.
8. Discuss the diffraction at a grating and obtain the condition for the mth maximum.
9. Discuss the experiment to determine the wavelength of monochromatic light using diffraction grating.
10. Discuss the experiment to determine the wavelength of different colours using diffraction grating.
11. Obtain the equation for resolving power of optical instruments.
12. Discuss about the simple microscope and obtain the equations for magnification for near point focusing and normal focusing.
13. Explain about compound microscope and obtain the equation for the magnification.
14. Obtain the equation for resolving power of microscope.
15. Discuss about astronomical telescope.
16. Mention different parts of spectrometer and explain the preliminary adjustments.
17. Explain the experimental determination of refractive index of the material of the prism using spectrometer.

Exercise problems

1. The ratio of maximum and minimum intensities in an interference pattern is 36 : 1. What is the ratio of the amplitudes of the two interfering waves?

   [Ans: 7:5]

2. In Young's double slit experiment, 62 fringes are seen on a screen for sodium light of wavelength 5893 Å. If violet light of wavelength 4359 Å is used

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